Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Hey Merlion!
b. Team leader name: Changhyeon Nam (team: Changhyeon Nam, Hyunho Yang)
c. Problem Statement: BFSI: Intelligent Risk, Fraud & Financial Experiences

Why this matters

THE PROBLEM

Banks are ready to lend on cash flow, but reading the records by hand costs too much, and the numbers a plain AI writes cannot be traced back to a page.

MEET LINH

Shopee seller, Vietnam

Asks for VND 150M of working capital. Her sales are real.

She brings 14 documents: a bank statement PDF, monthly Shopee dashboard screenshots, supplier receipts, photos of the shop.

No credit file. No collateral.
Today her file is declined, or priced blind.

Synthetic demo case B

1. **8.8%** of Vietnam's SMEs have a bank loan; large firms: 47.1%. Without a credit file or collateral, small firms are declined.
2. **75.5%** of Vietnamese firms cannot borrow without collateral, so banks are moving to cash-flow lending (VPBank, Vietcombank, Agribank).
3. **~45 min** to read and verify one file by hand. For a small loan, the reading costs more than the loan earns.
4. **77%** of the numbers in a memo written by a single Gemini prompt point to no page, so the underwriter has to find each one again. That prompt also caught only 15% of the planted risks.

Sources: 1 FiinGroup via Viet Nam News (Aug 2026) · 2 VCCI 2025 via VTV (May 2026) · 3 team estimate; ADB Asia SME Monitor 2025 · 4 our live benchmark, gemini-3.8-flash

Brief about the idea

THE SOLUTION

LedgerLens reads the shoebox, shows its work, and argues against itself before a human signs.

1 Read with evidence

ANSWERS: READING COST

Gemini reads every page. Each value carries document, page and bounding box. Python computes the cash flow. A memo with an unsourced number is rejected and regenerated.

In Linh's file: 14 documents, every figure sourced to a page and box, in about 2 minutes.

2 Challenge

ANSWERS: AI TRUST

A separate agent that never sees the writer's prompt runs deterministic checks and 12 underwriting questions, then ranks objections with evidence.

In Linh's file: Flags a reused receipt photo and VND 82.5M of own-account transfers counted as income.

3 Human signs

KEEPS: ACCOUNTABILITY

The underwriter accepts or dismisses each objection and signs. Every agent call is logged; decisions re-weight the next run.

In Linh's file: Linh gets the decision and the conditions to meet, in Vietnamese.

Grounded only in the applicant's documents. Guardrails: output schema, deterministic checks, a second agent, a human signature.

Opportunities

EXPECTED IMPACT

What changes, and for whom.

For the lender

45 → 2

minutes per file

USD ~11 → 0.28 per case. Small-ticket loans become worth underwriting.

For the small business

Cash flow

not collateral, decides the loan

Assessed on the records she already has. A no comes with reasons, in her own language.

For risk and audit

77% → 0%

of memo numbers with no page

Every figure points to a page and box, so nothing is re-checked by hand. Planted risks caught: 90% vs 15%.

USP The writer and the verifier are separate agents · numbers are computed by code, never generated · every number clicks back to its source line · decisions re-weight the checks

Back to the 8.8%: when reading a file costs cents and minutes instead of an hour, the loans banks decline today become worth making.

Manual baseline (45 min at USD 15/h) is a team estimate. A number counts as untraceable when no document and page stand behind it; correctness was not measured. Live gemini-3.8-flash on a small synthetic sample.

List of features offered by the solution

Every feature hangs on one pipeline, and the pipeline ends with a person.

Intake → Extract → Cash flow → Analyze → **Challenge** → Decide → Explain

Multimodal intake

PDF, screenshots, photos. Classified and quality-flagged.

Evidence-linked extraction

Doc id, page, bbox on every value. PDF text layer verifies it.

Deterministic model

Inflow, outflow, trend, volatility, debt service. Python, not the model.

Clickable memo

Cites catalog figures only. Click one, the source line opens.

Battery + 12 checks

Image hash, balance chain, dates. Ranked objections with evidence.

Underwriter workbench

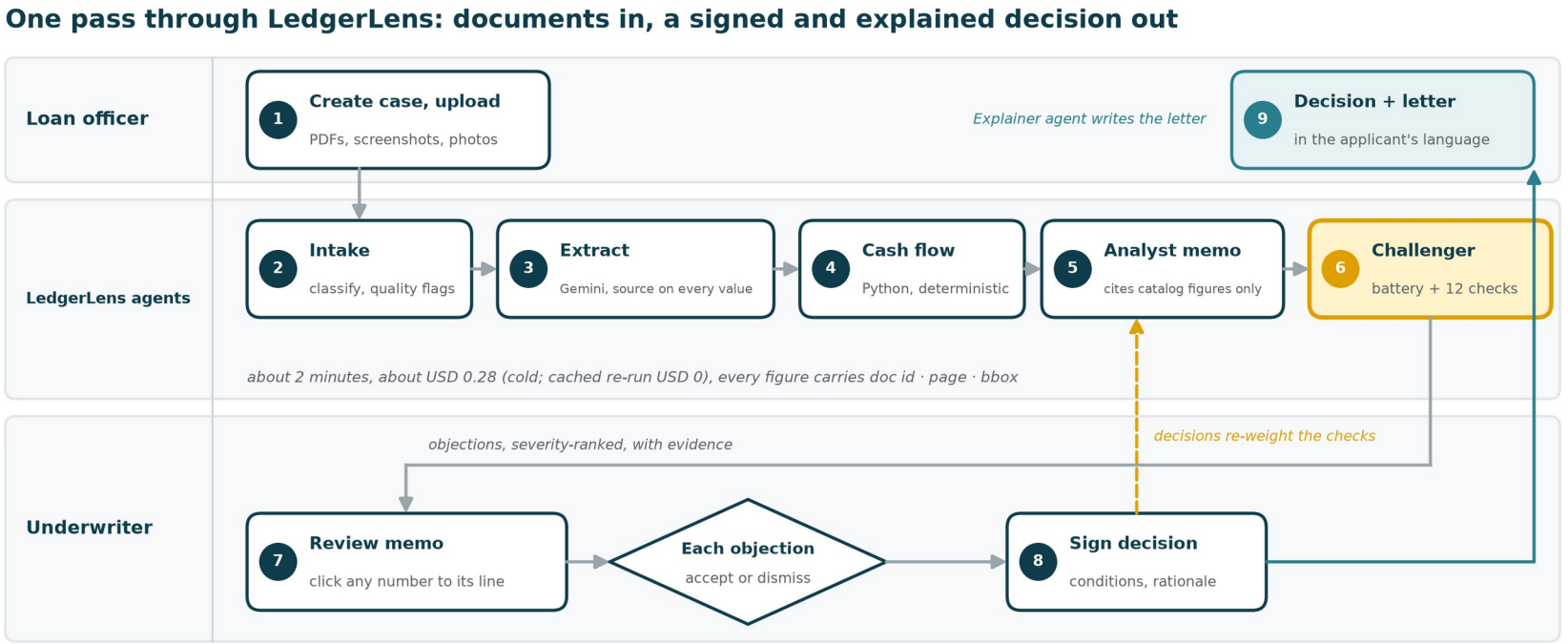
Accept or dismiss, sign. Audit log. Decisions re-weight the checks.

Letter and paper memo

Customer letter in the local language. Memo prints with footnoted sources.

Out of scope by design: credit scoring, loan pricing, core-banking integration. LedgerLens informs the decision; it does not make it.

Process flow diagram

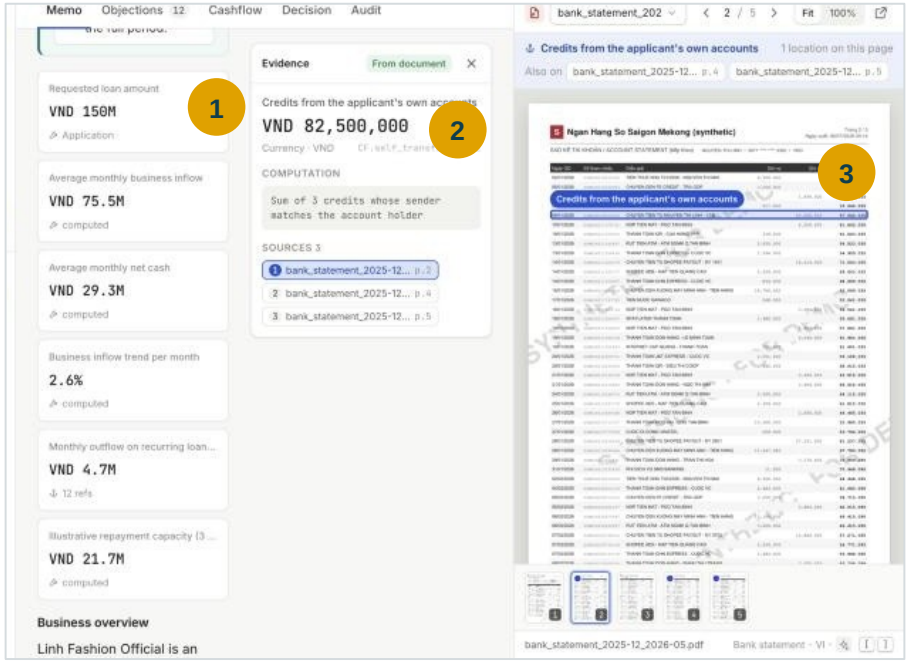


Three lanes: the loan officer uploads, the agents read and challenge, the underwriter decides. Decisions flow back into the Challenger's priorities.

Wireframes / Mock diagrams of the proposed solution

THE PRODUCT IN ONE INTERACTION

Click the number. See the line.



1 The memo cites a figure

Every underlined value is a catalog figure with a source. Here: VND 82,500,000, credits from the applicant's own accounts.

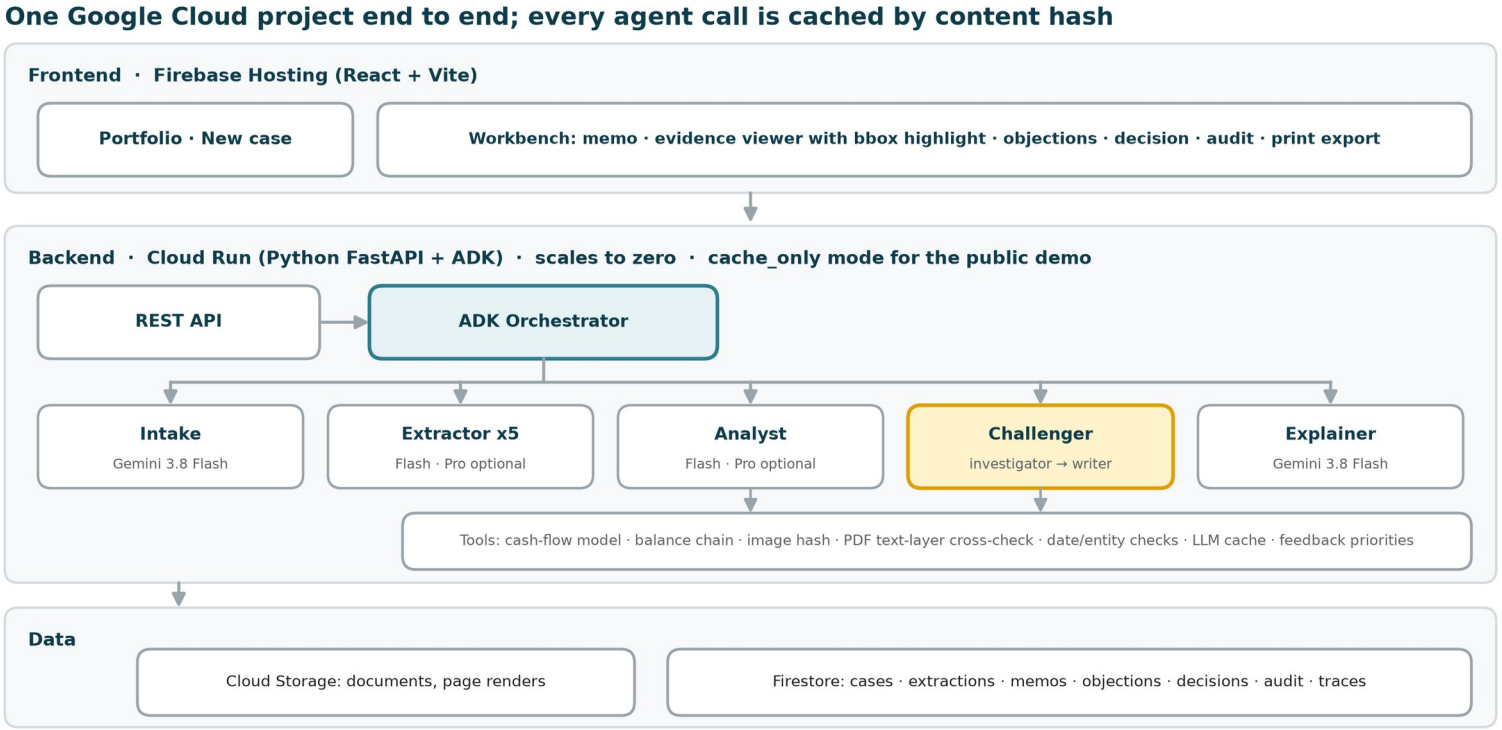
2 One click opens the evidence

The computation (sum of 3 credits whose sender matches the account holder) and all 3 source references: pages 2, 4 and 5.

3 The source line lights up

The statement page opens with the extractor's bounding box drawn on the exact row. A memo with an unsourced number is rejected and regenerated.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Gemini reads, ADK orchestrates, Python computes, Cloud Run serves.

GOOGLE CLOUD

Gemini 3.8 Flash

Default for every agent. Measured USD 0.26-0.31 per cold case; USD 0 cached.

GOOGLE CLOUD

Gemini 3.1 Pro

One env var moves Extractor, Analyst and Challenger to Pro.

GOOGLE CLOUD

ADK on Cloud Run

Agent Platform: Intake, 5 Extractors, Analyst, Challenger (investigator, writer), Explainer.

GOOGLE CLOUD

Cloud Run + Firebase

FastAPI backend scales to zero; React workbench on Firebase Hosting.

GOOGLE CLOUD

Firestore + Storage

Cases, extractions, memos, objections, decisions, audit, traces; documents.

PYTHON

Deterministic tools

Cash-flow model, balance chain, image hash, PDF text-layer cross-check, date and entity checks.

PYTHON

LLM cache, budget cap

Content-hash cache per model and prompt version; hard USD cap per process.

INTEGRATION

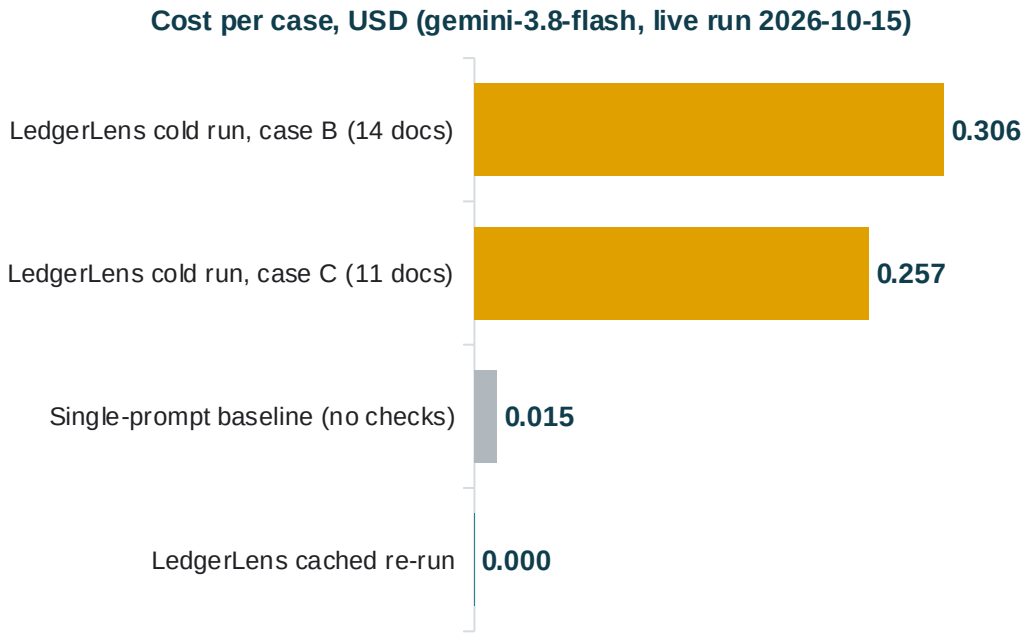
REST API

Any loan origination system can create cases, run stages and read memos.

Grounding: only the applicant's documents; every value must carry source {doc_id, page, bbox}. No separate OCR: Gemini reads rendered pages.

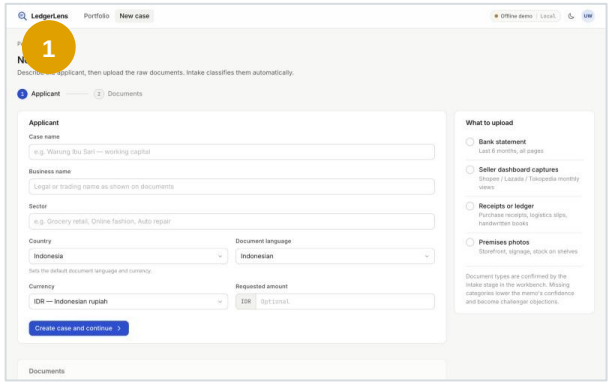
Estimated implementation cost

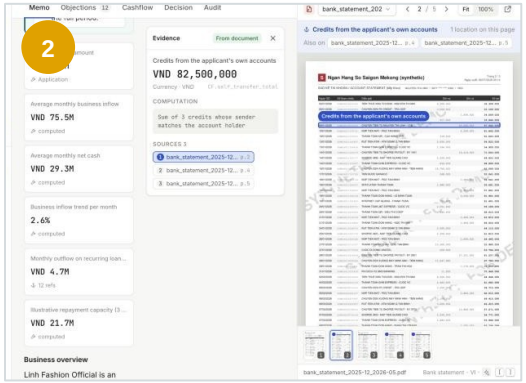
Measured: USD 0.26-0.31 per case on a cold Flash run, USD 0 on a cached re-run

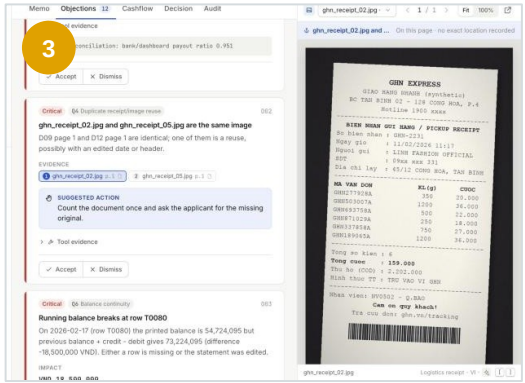


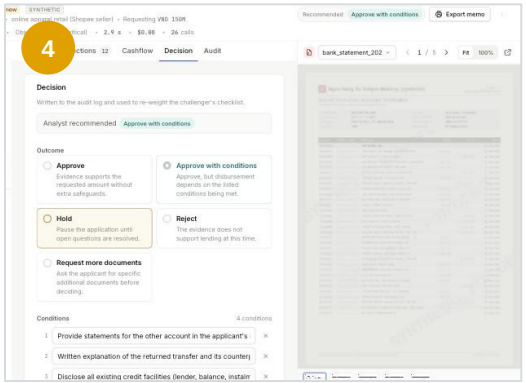
Assumption	Value
Gemini 3.8 Flash list price	USD 0.75 / 3.75 per 1M tokens
Measured cold run	~2 min and USD 0.26-0.31 per case
Cached re-run	USD 0 (content-hash LLM cache)
Public demo	Replays the recorded live Gemini run, USD 0; only the customer letter calls Gemini (~USD 0.0015)
Hosting	Cloud Run scales to zero; Firestore and Storage in free tier
Guardrail	Budget-cap env var stops Gemini calls once spent
Build	2 engineers, 4 days, free tier and trial credits

Snapshots of the prototype









1 New case

Applicant details, then the raw documents. Intake classifies and quality-flags them.

2 Click a number, see the line

Evidence panel with computation and all sources; the statement opens with the bounding box highlighted.

3 Challenger objections

Critical: two receipt photos are the same image with an edited date. Evidence chips open the photo.

4 The underwriter signs

Outcome, conditions and rationale, written to the audit log; accept/dismiss rates re-weight the checks.

Prototype performance report / Benchmarking

LedgerLens caught 90% of the planted risks on live Gemini output; a single prompt on the same documents caught 15%.

LEDGERLENS

90%

of planted risks caught

small synthetic sample, read as indicative

SINGLE PROMPT

15%

same documents

and same model

100%

of memo numbers trace to a page (single prompt: 23%)

~2 min

per case cold, USD 0.26-0.31; cached re-run USD 0

Metric	Target	Result
Risks caught (strict)	≥ 80%	90% (single prompt: 15%)
Traceable numbers	100%	100% (single prompt: 23%)
Objections on clean case A	≤ 20%	none
Extraction row accuracy*	≥ 97%	100%
Evidence-link accuracy*	≥ 95%	100%
Time per case (cold)	≤ 180 s	~2 min
Cost per case (cold)	≤ USD 0.50	USD 0.26-0.31

How measured. Live gemini-3.8-flash runs, 2026-10-15/16, on three synthetic businesses with team-planted risks. Strict scoring: an objection counts only if it carries the risk's identifying amount or document, one objection per risk. Traceable = a memo number with a document and page behind it; correctness not measured. The sample is small, so read every percentage as indicative. Missed: dashboard gross vs payout, COD vs cash deposits. Recommendation matched on C; A gave approve_with_conditions (expected approve), B request_more_documents (expected approve_with_conditions). *Clean born-digital synthetic PDFs; real scans will be lower.

Additional details / Future development

Where this goes next, and what we do not claim yet

Pilot Q1 2027

One SEA bank or MFI, real consented SME files; underwriter decisions replace synthetic traps

Integrate Q2–Q3 2027

Open-banking feeds (Brankas, Finverse) beside PDFs; hooks into loan origination systems

Scale 2028

Same pattern for trade finance and micro-insurance; shared Challenger check library

What we do not claim yet, and which phase answers it

Synthetic benchmark

A small synthetic sample with planted risks; real-data results come with the pilot.

Forged documents

Hash and balance-chain checks catch crude edits. Open-banking data closes the gap for the rest.

Platform sellers

Shopee and Grab already lend to their own sellers. We target banks and MFIs lending outside platforms.

positioning, not a feature

Humans decide, every figure and decision is traceable, the prototype uses watermarked synthetic data only, and production keeps documents and model endpoints in-region.

Provide links to your:

GitHub public repository

github.com/changhyeonnam/ghack

English README, architecture, run instructions, synthetic-data notice, presentation/

Demo video (3 to 4 minutes)

TBD

Problem in 20 seconds, click-to-source and Challenger scenes, benchmark near the end

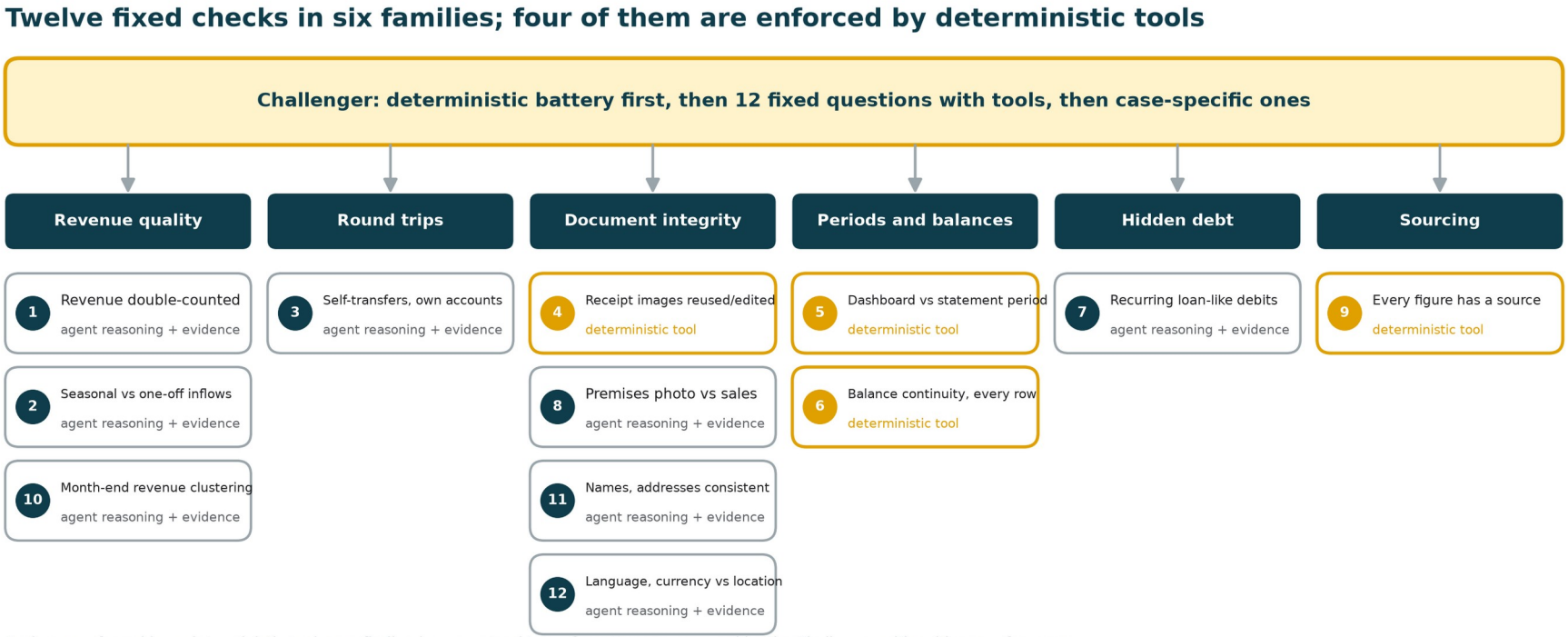
Final product

TBD

Cloud Run service; each browser gets its own copy of the three demo cases (Reset demo). Scales to zero; demo cases replay the recorded live Gemini run (only the customer letter calls Gemini live), so judging costs nothing

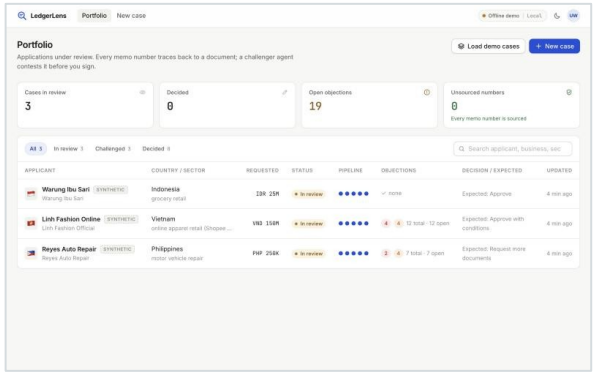
Links are inserted at submission on 18 October 2026.

Appendix: the Challenger's 12 fixed checks



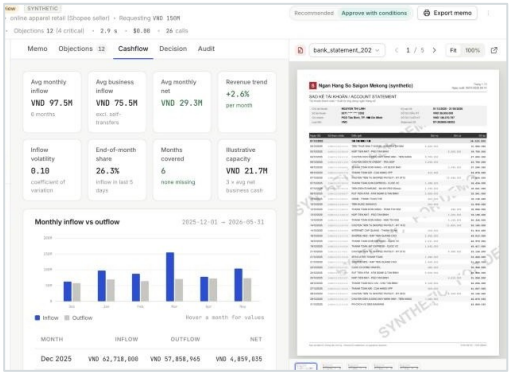
Amber = enforced by a deterministic tool, so a finding is guaranteed to surface. Grey = reasoned by the Challenger with evidence references.

Appendix: portfolio, cash flow and audit views



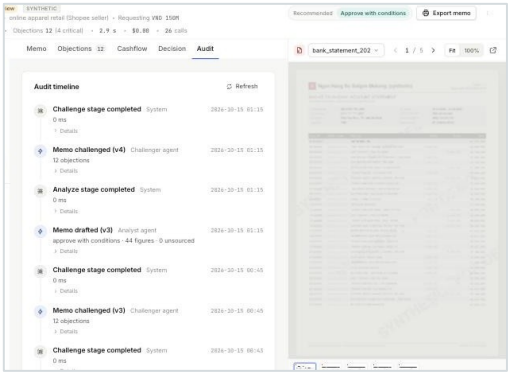
Portfolio

Cases in review, open objections and unsourced numbers across the book; filters by status.



Cash flow

Deterministic monthly model: inflow, net, trend, volatility, end-of-month share, repayment capacity.



Audit timeline

Every stage and agent call with version, model and prompt; the trail a regulator can read.

Also in the build: dark and light themes, PDF text-layer cross-check on every extracted amount, content-hash LLM cache, cache_only demo mode, budget cap, printable paper memo with footnoted sources.

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AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

